

THE CURVE OF TRUTH



The data is a chaotic sea. Somewhere within the noise lies the signal... the life-saving truth. I just need the right tool to separate them.

Mastering the ROC:
An Epidemiological Manga Story

PATIENT ID: 7044 // BIOMARKER:
Total Activity Count (TAC) //

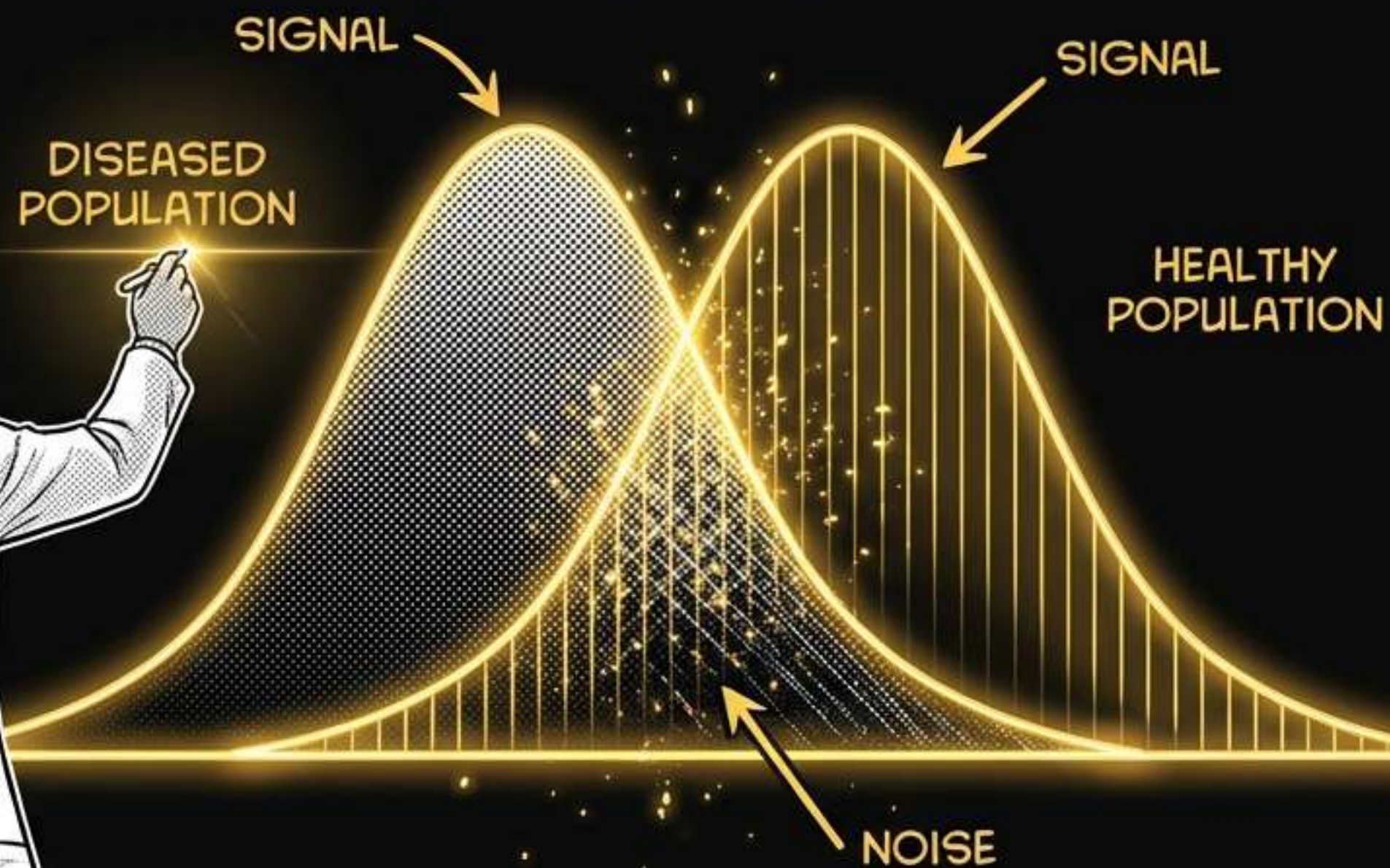
VALUE:

600

You are looking for
a binary answer in a
continuous world,
Amina. Life is rarely
so black and white.

It's 600. Not
high enough to be
safe... not low enough to
confirm mortality risk.
in the Gray Zone. How
do I classify this?

Observe the overlap.
Some healthy patients have
low scores; some sick
patients have high scores.
No test is perfect.



True Positive (TP)



False Positive (FP)



False Negative (FN)



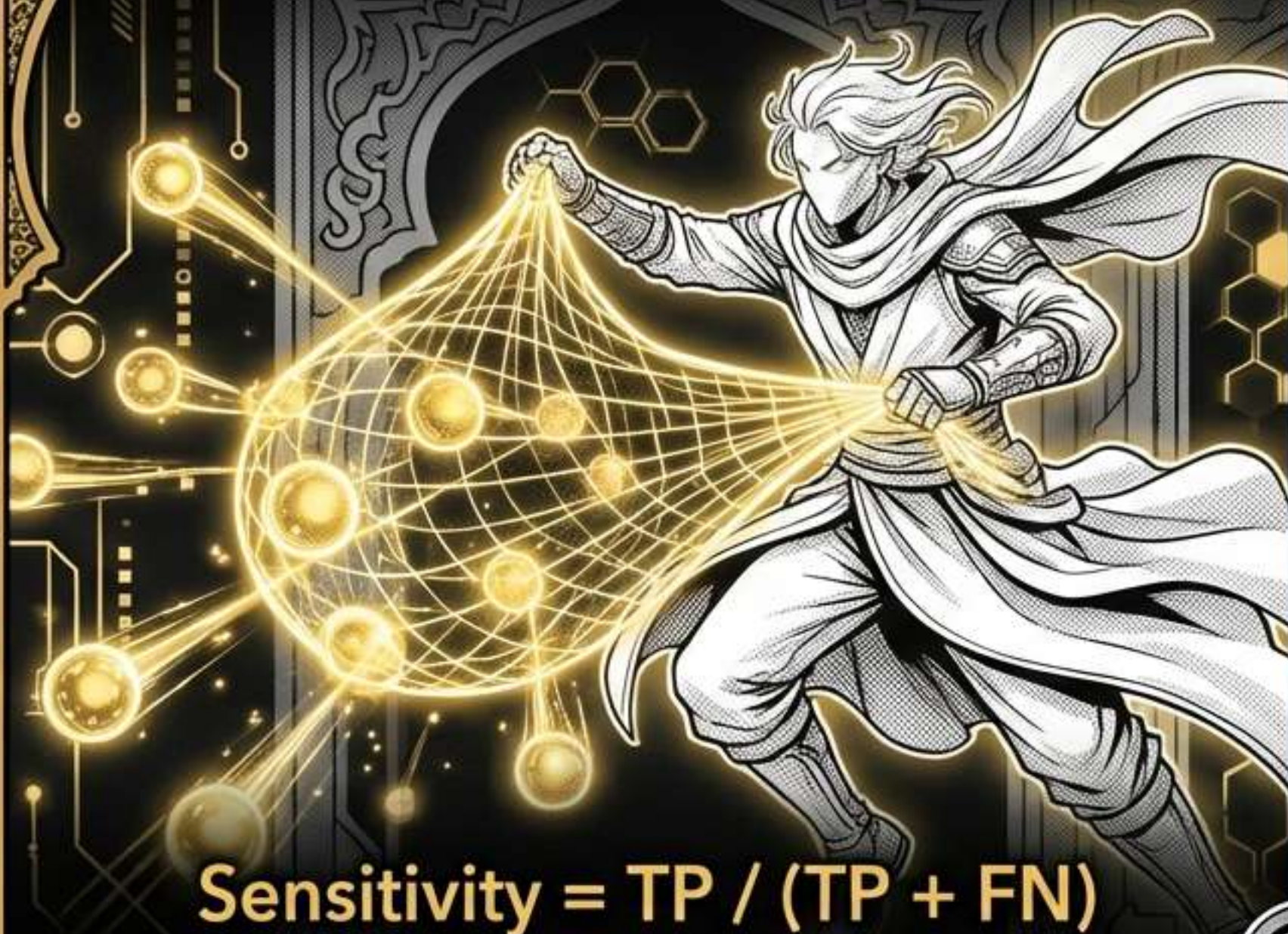
True Negative (TN)



Every decision has a cost.
A False Alarm causes stress.
A False Negative... costs a life.
You must balance them.



SENSITIVITY



$$\text{Sensitivity} = \text{TP} / (\text{TP} + \text{FN})$$

The ability to catch the disease.

SPECIFICITY



$$\text{Specificity} = \text{TN} / (\text{TN} + \text{FP})$$

The ability to clear the healthy.

They are like twin forces...
if I strengthen the Hunter, the
Shield weakens?



Dr. Rayan:

The Cut-off Value is your blade.
Where you place it determines
who you save and who you
misdiagnose.

****CUT-OFF VALUE****

Healthy
Diseased

Strict Cut-off ->
High False Positives

Lenient Cut-off ->
High False Negatives.

The Trade-off: Moving the threshold changes
the balance of Sensitivity and Specificity.

PERFECTION

**Sensitivity
(True Positive Rate)**

(0,0)

**Sensitivity
(True Positive Rate)**

(1,1)

(1,1)

**1 - Specificity (False Positive Rate)
(Komika Axis)**

Welcome to the
ROC Space. A legacy of
of radar operators, now our
map for medical truth.
Forget the distributions.
Map the relationship.



Empirical ROC
(Non-Parametric)
in Komika Axis

Binormal ROC
(Parametric)
in Komika Axis

True Positive Rate (Sensitivity)

False Positive Rate (1-Specificity)
(in Komika Axis)

The Empirical is raw...
but the Parametric is smooth.
It assumes the data follows
a normal distribution.
in CC Wild Words.

$AUC = 0.5 = \text{FAILURE}$

Beware the Diagonal.
This is the line of the coin toss.
If your curve touches this,
your science is no better
than gambling.

True Positive Rate

0

The Line of Chance ($AUC = 0.5$)

False Positive Rate (1-Specificity)
(Komika Axis)

1

AUC (Area Under the Curve)

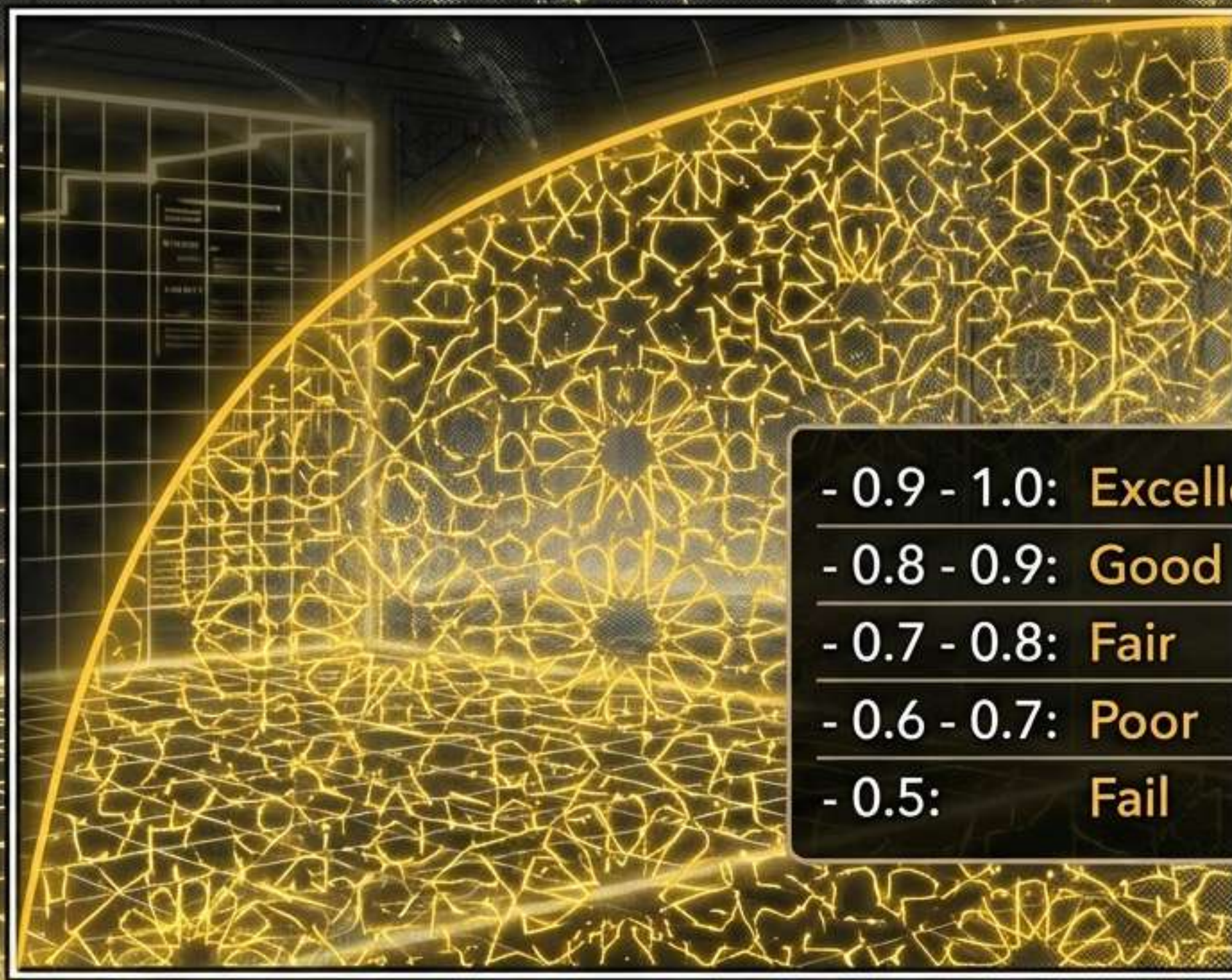


It measures Discrimination...
The probability that a sick
patient scores higher than
a healthy one.

True Positive Rate (Sensitivity)

1

0



- 0.9 - 1.0: **Excellent**
- 0.8 - 0.9: **Good**
- 0.7 - 0.8: **Fair**
- 0.6 - 0.7: **Poor**
- 0.5: **Fail**

False Positive Rate (1-Specificity)
(Komika Axis)

1

مَنَّا الْقِيَان

The curve shows Potential.
But I need a Strategy.
Where do I plant the flag
for Patient 7044?

True Positive Rate (Sensitivity)

DECISION



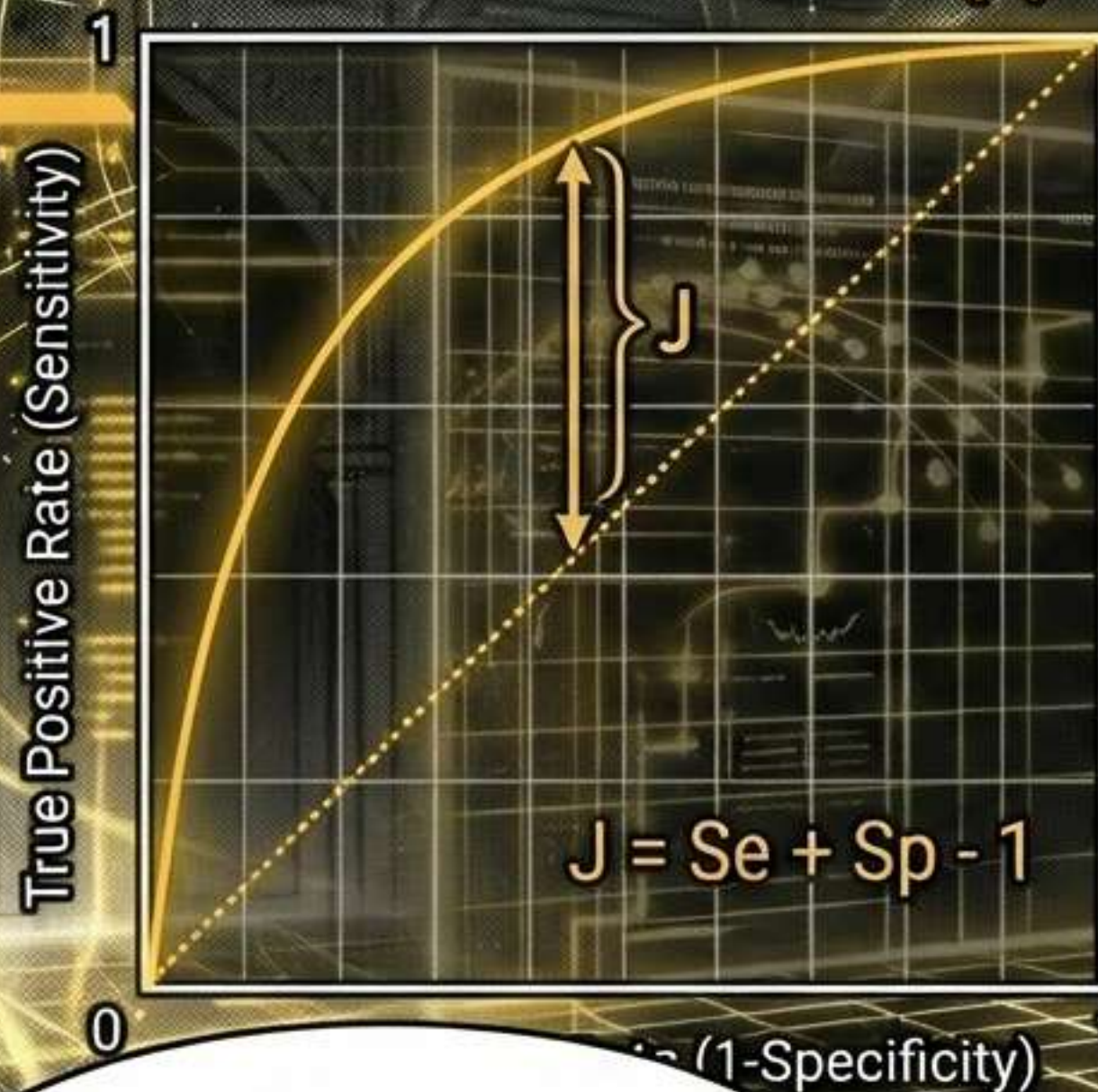
Optimization Strategies:

Avenir Next

- 1. Youden's Index
- 2. Euclidean Distance
- 3. Cost Approach

False Positive Rate (1-Specificity)
(Komika Axis)

YOUDEN'S INDEX (J)



Youden seeks the point farthest from ignorance. Euclidean seeks the point closest to perfection.

EUCLIDEAN DISTANCE



Youden seeks the point farthest from ignorance. Euclidean seeks the point closest to perfection.



Cost Index Approach

$$\text{Cost Index Slope} = \left(\frac{(1 - \text{Prev})}{\text{Prev}} \right) * \frac{(\text{CostFP} - \text{CostTN})}{(\text{CostFN} - \text{CostTP})}$$

****Cost of
Missed
Diagnosis**
(FN)**

****Cost of
False Alarm
(FP)**

Context is king.
If the disease is fatal, we
accept more False Positives.
Mathematics must bow
to reality.

Dataset: NHANES (2011-2014). N=5,006.
Biomarker: Total Activity Count.
Outcome: Mortality.

****MISCLASSIFICATION
DETECTED****

Sensei!
I used the optimal cut-off...
but the predictions for the
elderly patients are wrong!
The curve is lying!

The curve is not lying.
You are ignoring the
Context.

THE COVARIATE PROBLEM

YOUTH



High Activity = Normal

ELDERLY

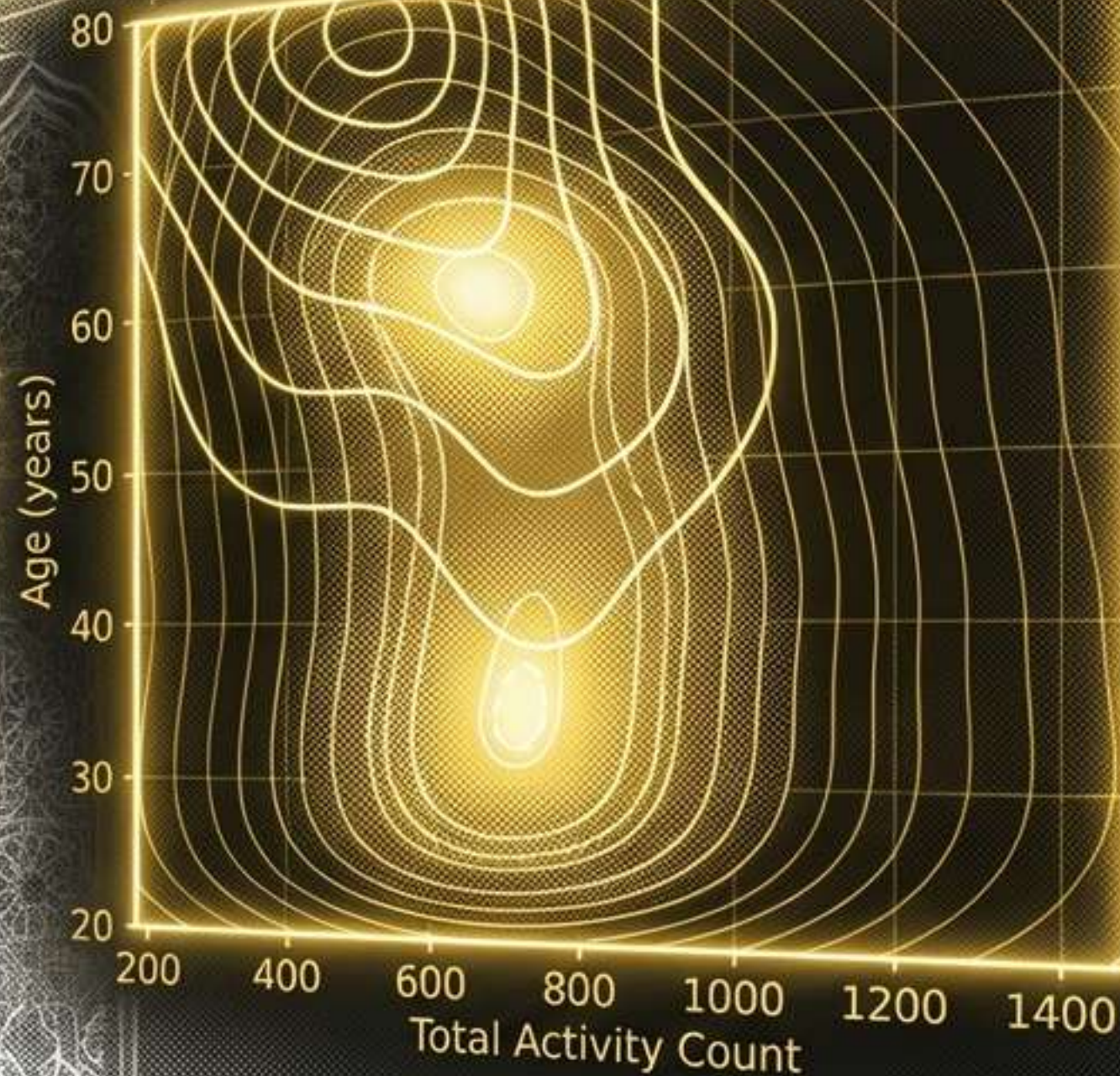


Low Activity = Normal

A "low" score for a 20-year-old is a "heroic" score for an 80-year-old. You cannot judge them by the same threshold. We need **Adjustment**.



The distribution shifts... and warps! It's non-linear. A simple straight line won't fit this.
CC Wild Words.



"Visualizing Heterogeneity: The Signal changes shape with Age." in Avenir Next.

METHOD: AROC via NEURAL NETWORKS

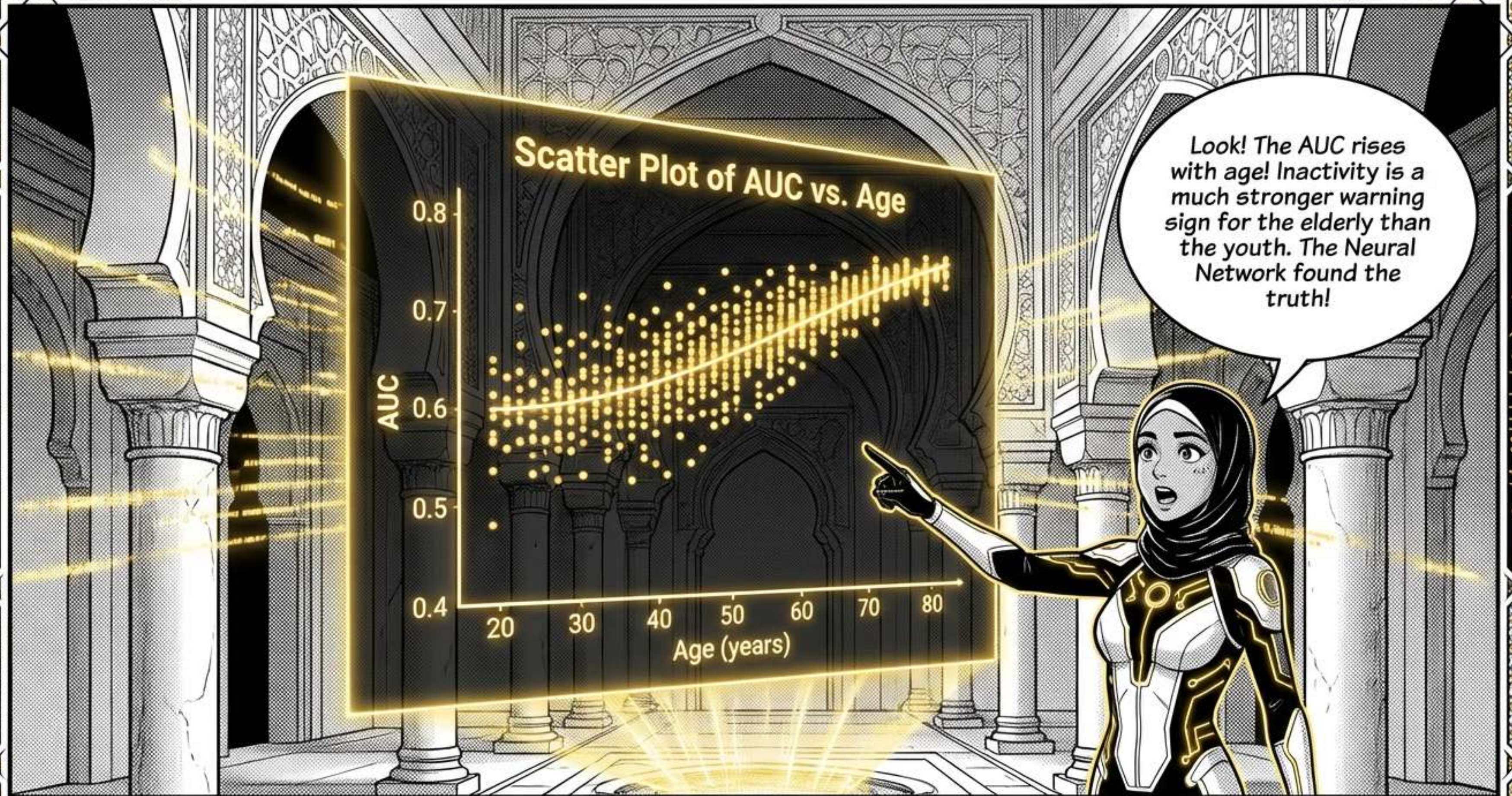
Stage 1: Estimate Conditional Mean $\mu(x)$

Stage 2: Estimate Conditional Variance $\sigma^2(x)$

Technique: Multi-Layer Perceptron (MLP)
with ReLU Activation.



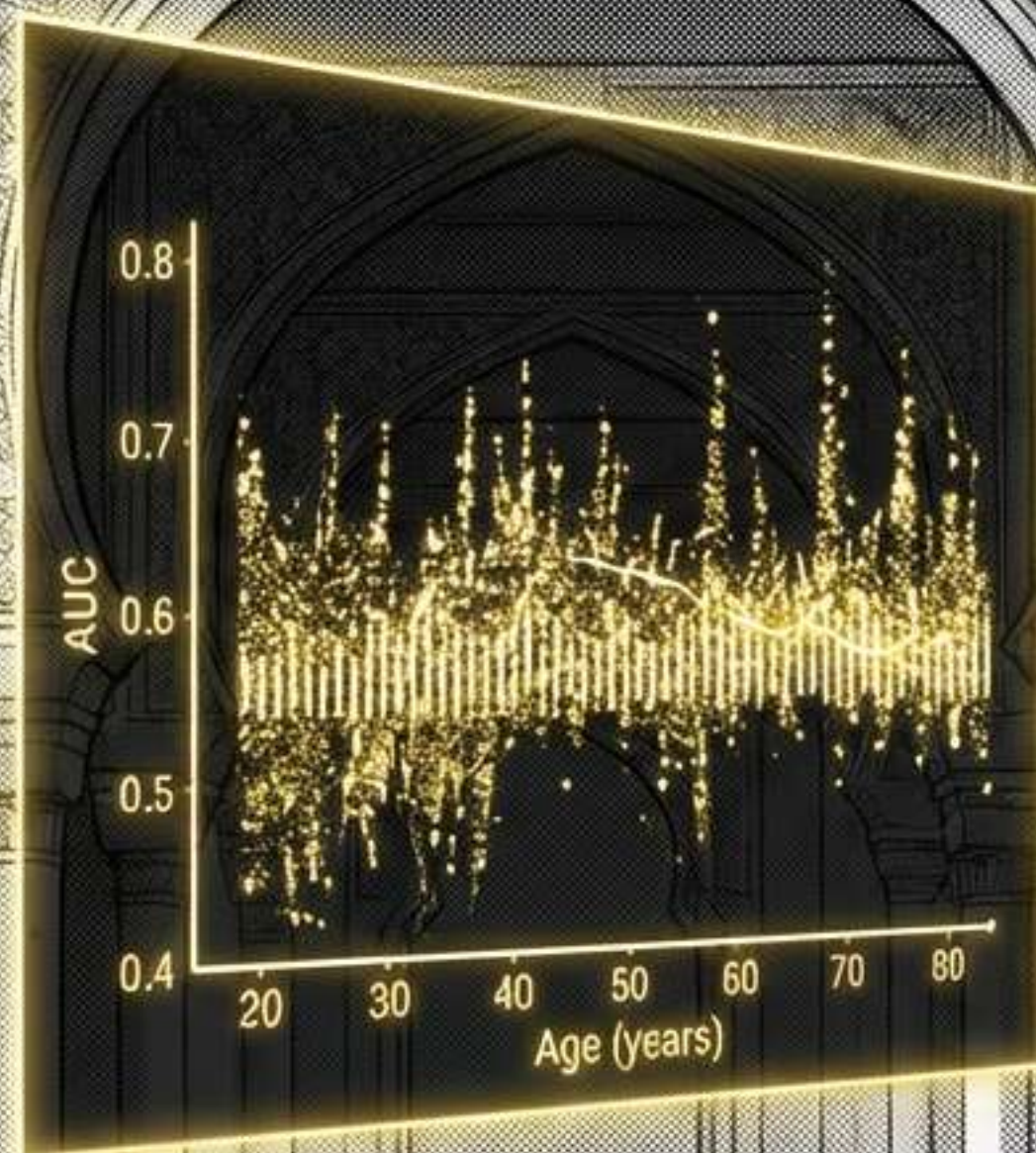
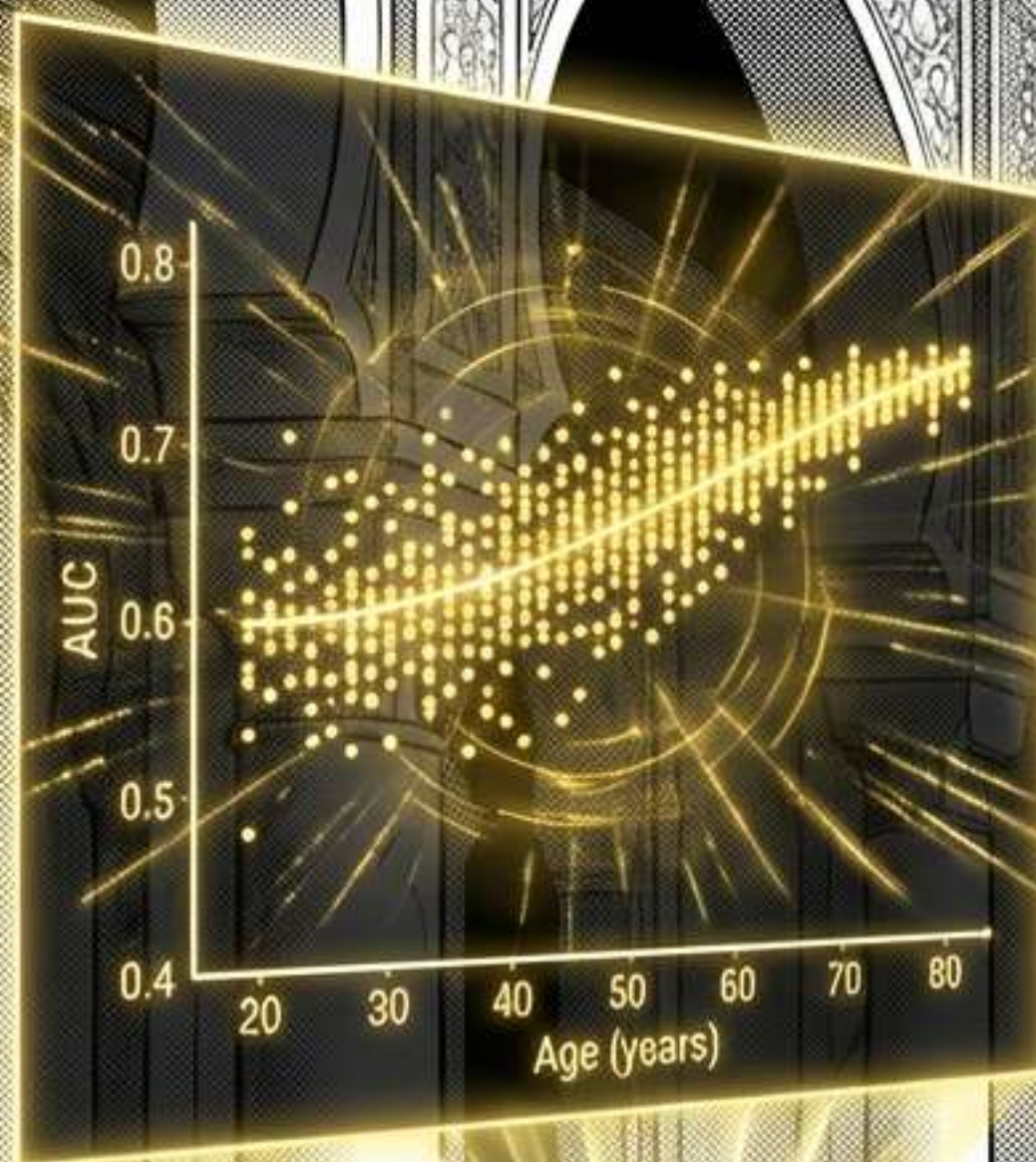
We will train the
Neural Network to learn
the curves **within**
the curves!



FNN Model

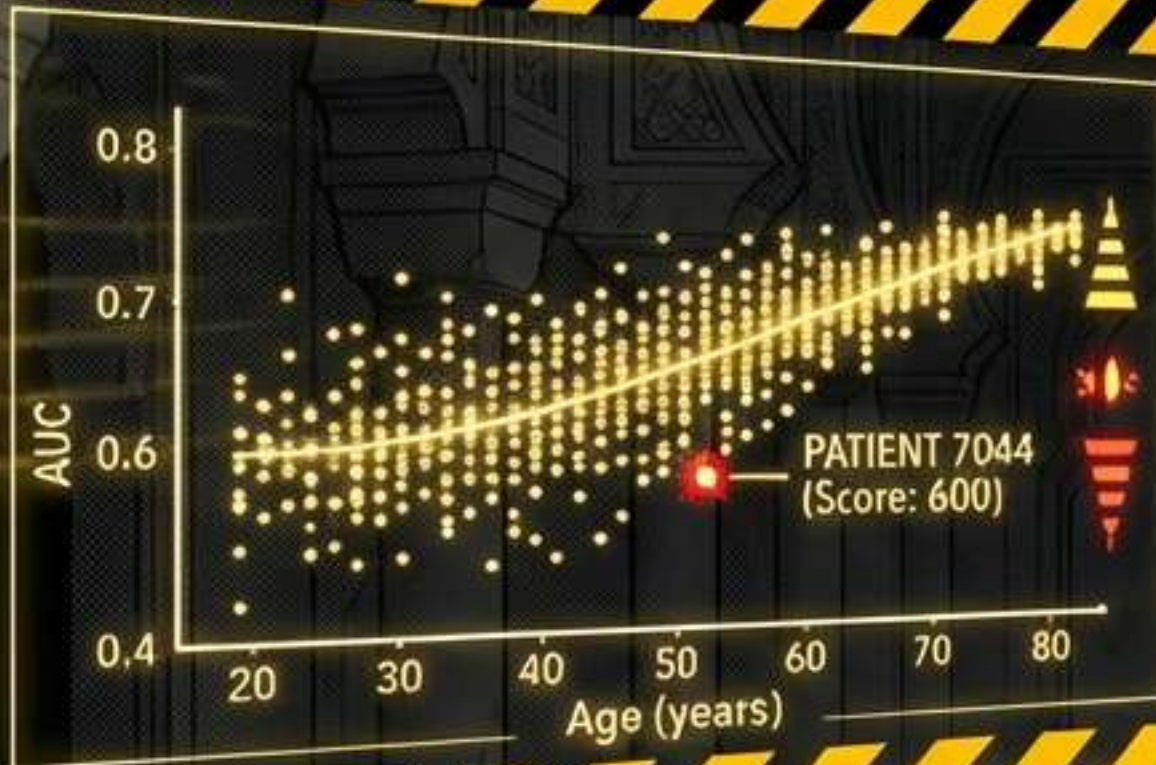
Random Forest

Semiparametric



The Neural Network captures the smooth, non-linear reality that other models miss.

**! ADJUSTED THRESHOLD APPLIED.
PATIENT 7044 SCORE (600) IS
BELOW EXPECTATION FOR AGE 70.
>>> HIGH RISK DETECTED.**



DATA FILTER

AGE: 70

Patient 7044 is 70 years old. Adjusted for his age, this score is a clear warning. Intervention Recommended.

1. ROC: Maps Sensitivity vs. 1-Specificity.

2. AUC: The measure of discrimination (1.0 = Perfect).

3. Youden/Cost: Strategies for the Optimal Cut-off.

4. AROC/Neural Nets: The future of personalized accuracy.

You have learned well.
Data without context is noise.
Data *with* context... is truth.

KNOWLEDGE IS THE SEPARATION OF TRUTH FROM ERROR.